

EENG 350 Digital Signal Processing

Spring 2026

Homework 6

Note: You must include your reasoning and calculation procedure to receive full credit.

1. Determine the DTFT of $x[n] = n\alpha^n \mu[n+2]$, $|\alpha| < 1$ ($\mu[n]$ is unit step function). (3pts)
2. Determine the DTFT of $y[n] = \mu[n+2] - \mu[n-3]$ ($\mu[n]$ is unit step function). (3 pts)
3. Let $X(e^{j\omega})$ denote the DTFT of a length-8 sequence $x[n]$ given by
 $x[n] = \{2, 3, -1, 0, -4, 3, 1, 2\}$, $-2 \leq n \leq 6$
Evaluate the following functions of $X(e^{j\omega})$ **without** computing the transfer function itself.
(10 pts)

(a) $X(e^{j0})$

(b) $X(e^{j\pi})$

(c) $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$

(d) $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

(e) $\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$